

संभव BATCH - ANSWER KEY

Geography: Indian Drainage System & Ocean Currents (Moderate Level)

Under the guidance of Vipulendra Tripathi Sir

Detailed Solutions & Explanations (Zero Error Verified)

Question 1

Ans: (D) All I, II, and III

Explanation: The Gulf Stream is a warm current originating in the Gulf of Mexico. The Labrador Current is a cold current flowing from the Arctic. Their confluence near Newfoundland creates dense fog but also stirs up nutrients, making it one of the world's richest fishing grounds (Grand Banks).

Question 2

Ans: (C) The Agulhas Current is a cold current in the Indian Ocean.

Explanation: The Agulhas Current is actually a warm western boundary current in the southwestern Indian Ocean, flowing down the east coast of Africa. The other statements correctly describe general ocean current dynamics.

Question 3

Ans: (A) a-2, b-3, c-4, d-1

Explanation: Kuroshio is a warm current in the Pacific. Oyashio is a cold current in the Pacific. Canary is a cold current in the North Atlantic. Benguela is a cold current in the South Atlantic.

Question 4

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: Cold currents (like the Benguela, Peru/Humboldt, and Canary) flow along the western coasts of continents in subtropical latitudes. These cold currents chill the air above them, reducing its moisture-holding capacity and preventing rainfall, leading directly to the formation of coastal deserts like the Namib and Atacama.

Question 5

Ans: (B) Peru Current

Explanation: The Humboldt Current is a cold, low-salinity ocean current that flows north along the western coast of South America, prominently along Peru and Chile, hence widely known as the Peru Current.

Question 6

Ans: (A) Only I and II

Explanation: The Sargasso Sea is located in the North Atlantic Gyre. It is famous for its distinct brown Sargassum seaweed. However, its waters are famously calm and sluggish, not turbulent, because it lies in the eye of a large ocean gyre.

Question 7

Ans: (C) Falkland Current - Indian Ocean

Explanation: The Falkland Current is a cold ocean current that flows northward along the Atlantic coast of Patagonia (South America). It is in the Atlantic Ocean, not the Indian Ocean.

Question 8

Ans: (A) a-2, b-1, c-4, d-3

Explanation: Brazil Current flows in the South Atlantic. Alaska Current flows in the North Pacific. West Australian Current flows in the Indian Ocean. North Atlantic Drift flows in the North Atlantic Ocean.

Question 9

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: When warm air over a warm current meets the cold air over a cold current, condensation occurs, producing thick, persistent fog. This happens at the Grand Banks where the warm Gulf Stream meets the cold Labrador Current, creating ideal conditions for plankton and fish, despite the hazardous fog.

Question 10**Ans: (C) Kuroshio Current**

Explanation: The Kuroshio Current is a strong surface oceanic current of the Pacific Ocean, carrying warm water from the equator towards the north. Canary, Benguela, and Oyashio are all cold currents.

Question 11**Ans: (A) Only I and II**

Explanation: El Nino warms the eastern Pacific and brings rainfall to the usually dry Peruvian coast. However, statement III is incorrect because El Nino typically weakens the Indian monsoon, causing drought-like conditions in the subcontinent, not strengthening it.

Question 12**Ans: (C) It flows in the same direction as the Trade Winds.**

Explanation: The Equatorial Counter Current flows eastward (west to east), while the Trade Winds blow westward (east to west). Therefore, it flows opposite to, not in the same direction as, the Trade Winds.

Question 13**Ans: (A) a-3, b-4, c-1, d-2**

Explanation: Gulf Stream is a warm current (a-3). Labrador is a cold current (b-4). Agulhas is a current in the Indian Ocean (c-1). Kamchatka is a current in the Pacific Ocean (d-2).

Question 14**Ans: (A) Both A and R are true and R is the correct explanation of A**

Explanation: Ocean currents are critical for regulating global climate. They distribute heat by moving warm equatorial water towards the higher latitudes and cold polar water towards the equator, effectively balancing the Earth's thermal budget.

Question 15**Ans: (C) Somali Current**

Explanation: The Somali Current in the Indian Ocean undergoes a complete seasonal reversal. During the summer, the strong South-West Monsoon drives it northwards. During the winter, the North-East Monsoon reverses its flow to the south.

Question 16**Ans: (A) I, IV, III, II**

Explanation: Tracing the North Atlantic Gyre clockwise: It begins with the North Equatorial Current (I) flowing west, turns north as the Gulf Stream (IV), crosses the ocean eastwards as the North Atlantic Drift (III), and flows south as the Canary Current (II) to complete the loop.

Question 17**Ans: (C) They increase the moisture holding capacity of the air, causing heavy rainfall.**

Explanation: Cold currents lower the temperature of the air above them. Cold air holds less moisture, leading to stability and lack of condensation. Consequently, they cause aridity and desertification on coastal areas, rather than heavy rainfall.

Question 18**Ans: (A) a-3, b-4, c-1, d-2**

Explanation: The Atacama Desert is associated with the Peru/Humboldt Current. The Namib Desert is tied to the Benguela Current. The Great Victoria Desert correlates with the West Australian Current. The Mojave Desert is associated with the California Current.

Question 19**Ans: (A) Both A and R are true and R is the correct explanation of A**

Explanation: The North Atlantic Drift (a continuation of the warm Gulf Stream) carries immense heat to the shores of Western Europe. This keeps the coastal waters and ports (like those in the UK and Norway) ice-free year-round, despite their high latitudes.

Question 20

Ans: (A) Benguela Current

Explanation: The Benguela Current is a broad, northward flowing cold ocean current that forms the eastern portion of the South Atlantic Ocean gyre.

Question 21

Ans: (A) Only I and II

Explanation: La Nina brings cooler-than-average sea surface temperatures to the central and eastern Pacific. This typically results in a stronger monsoon in India. It brings heavy rains and floods to Australia, not drought-like conditions (El Nino brings drought to Australia).

Question 22

Ans: (C) Irminger Current

Explanation: The Irminger Current is a North Atlantic ocean current flowing westward off the southwest coast of Iceland. The East Australian, Kuroshio, and Oyashio currents all belong to the Pacific Ocean.

Question 23

Ans: (B) Temperature and salinity

Explanation: Thermohaline circulation (thermo = temperature, haline = salinity) is the deep-ocean circulation driven by large-scale variations in water density, which are dictated by temperature and salt content.

Question 24

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: Upwelling occurs when winds blow surface water away from a coast, causing deep, cold, nutrient-rich water to rise to the surface. These nutrients fertilize the surface waters, leading to massive phytoplankton blooms that support massive fish populations.

Question 25**Ans: (B) Canary Current**

Explanation: The Sargasso Sea in the North Atlantic is bounded on the west by the Gulf Stream, on the north by the North Atlantic Drift, on the east by the Canary Current, and on the south by the North Equatorial Current.

Question 26**Ans: (A) Only I and II**

Explanation: Himalayan rivers are snow-fed and perennial. Peninsular rivers rely on rain and are largely seasonal. Statement III is incorrect because the Narmada and Tapi flow westwards into the Arabian Sea, not eastwards into the Bay of Bengal.

Question 27**Ans: (C) The Yamuna joins the Ganga at Varanasi.**

Explanation: The Yamuna is the longest tributary of the Ganga, but it joins the Ganga at Triveni Sangam in Prayagraj (formerly Allahabad), not at Varanasi.

Question 28**Ans: (A) a-2, b-4, c-1, d-3**

Explanation: Godavari originates at Trimbakeshwar (Nashik). Krishna rises at Mahabaleshwar. Narmada originates from the Amarkantak Plateau. Mahanadi rises from Sihawa in Chhattisgarh.

Question 29**Ans: (A) Both A and R are true and R is the correct explanation of A**

Explanation: Narmada and Tapi flow through hard rock rift valleys with high gradients. Due to their rapid velocity and lack of heavy sediment deposition at their mouths, ocean tides easily wash away whatever silt is present, leaving funnel-shaped estuaries instead of deltas.

Question 30**Ans: (C) Godavari**

Explanation: Due to its large size and extent—second only to the Ganga in India—the Godavari river is traditionally known as the Dakshin Ganga (Ganges of the South).

Question 31**Ans: (B) Krishna**

Explanation: The Tungabhadra, Bhima, Musi, Koyna, and Ghataprabha are all major tributaries of the Krishna River, one of the longest rivers in central-southern India.

Question 32**Ans: (D) It forms the Sundarbans Delta exclusively with the Mahanadi river.**

Explanation: The Brahmaputra merges with the Ganga (and the Meghna in Bangladesh) to form the massive Sundarbans Delta. It has no connection with the Mahanadi river, which forms its own delta further south in Odisha.

Question 33**Ans: (A) a-3, b-1, c-4, d-2**

Explanation: Manjira is a major right-bank tributary of the Godavari. Betwa joins the Yamuna. Indravati is a crucial left-bank tributary of the Godavari. Subansiri is the largest tributary of the Brahmaputra.

Question 34**Ans: (A) Both A and R are true and R is the correct explanation of A**

Explanation: The Kosi river descends from Nepal carrying excessive silt. This causes the riverbed to choke and forces the river to frequently shift its channel (braiding), leading to catastrophic floods in the flat plains of Bihar.

Question 35**Ans: (C) Rann of Kutch**

Explanation: The Luni River originates in the Aravalli range and flows southwest. Because of the arid climate and high evaporation, it fails to reach the sea and terminates into the saline marshes of the Rann of Kutch (inland drainage).

Question 36**Ans: (D) All I, II, and III**

Explanation: Damodar, Subarnarekha, and Baitarani are all major east-flowing peninsular rivers. They empty into the Bay of Bengal, traversing through states like Jharkhand, West Bengal, and Odisha.

Question 37**Ans: (C) Vijayawada - Godavari**

Explanation: Vijayawada is situated prominently on the banks of the Krishna River, not the Godavari. The Godavari flows through cities like Rajahmundry.

Question 38**Ans: (A) a-2, b-3, c-4, d-1**

Explanation: At Rudraprayag, Alaknanda meets Mandakini. At Karnaprayag, it meets Pindar. At Vishnuprayag, it meets Dhauliganga. At Devprayag, the Alaknanda meets the Bhagirathi to finally become the Ganga.

Question 39**Ans: (A) Both A and R are true and R is the correct explanation of A**

Explanation: Himalayan rivers (Ganga, Brahmaputra) have massive, young, and active catchment areas. They perform intense erosion and carry colossal volumes of silt, which settle at their mouths in the flat plains, forming the world's largest deltas (like the Sundarbans).

Question 40**Ans: (C) Mahi**

Explanation: The Mahi River in western India originates in Madhya Pradesh, flows northwards crossing the Tropic of Cancer, then turns southwards and crosses the Tropic of Cancer a second time before entering Gujarat.

Question 41**Ans: (C) Both I and II**

Explanation: The 1960 Indus Water Treaty, brokered by the World Bank, gave India control over the three eastern rivers (Ravi, Beas, Sutlej). Pakistan gained control over the three western rivers (Indus, Jhelum, Chenab), with India permitted limited agricultural and domestic uses.

Question 42**Ans: (C) Sambhar Lake**

Explanation: Wular (J&K), Loktak (Manipur), and Kolleru (AP) are prominent freshwater lakes. Sambhar Lake in Rajasthan is India's largest inland salt (saline) lake.

Question 43**Ans: (A) a-3, b-2, c-1, d-4**

Explanation: Hirakud Dam is built on the Mahanadi River. Bhakra Nangal is constructed on the Sutlej. Sardar Sarovar is on the Narmada River. Nagarjuna Sagar is built across the Krishna River.

Question 44**Ans: (A) Both A and R are true and R is the correct explanation of A**

Explanation: The Cauvery river is unique among peninsular rivers because its upper catchment (in Karnataka) receives rain from the advancing SW monsoon, and its lower catchment (in Tamil Nadu) receives rain from the retreating NE monsoon, keeping it fed almost year-round.

Question 45

Ans: (A) Narmada and Son

Explanation: The Amarkantak Plateau in Madhya Pradesh is a unique radial drainage center. The Narmada River originates here and flows westward, while the Son River originates nearby and flows northward to join the Ganga.

Question 46

Ans: (D) All I, II, and III

Explanation: All three pairs are accurate. Jog Falls is created by the Sharavati River. Hundru Falls is on the Subarnarekha River near Ranchi. Sivasamudram Falls is formed by the Cauvery River.

Question 47

Ans: (C) Pennar

Explanation: The Pennar river flows eastwards through Karnataka and Andhra Pradesh into the Bay of Bengal. Mandovi, Zuari, and Periyar are all west-flowing rivers draining into the Arabian Sea.

Question 48

Ans: (B) Brahmaputra

Explanation: Majuli is a massive riverine island located in Assam. It is formed entirely by the Brahmaputra River and its tributaries, and holds the distinction of being India's first island district.

Question 49

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: The Peninsular landmass is a relic of the ancient Gondwanaland. As a result, its rivers have been eroding the landscape for a much longer time, reaching their mature stage (base level of erosion), characterized by broad, shallow, graded valleys.

Question 50

Ans: (D) All I, II, III, and IV

Explanation: The Yamuna receives several significant right-bank tributaries originating from the Peninsular plateau, including the Chambal, Sind, Betwa, and Ken, before it eventually joins the Ganga.